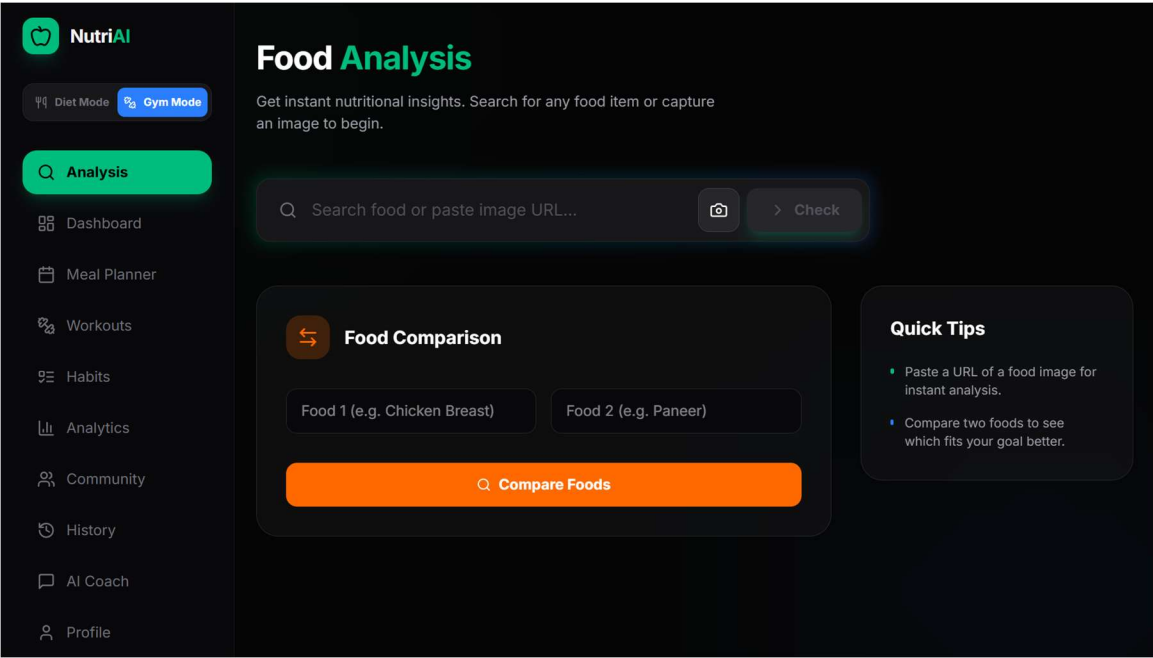


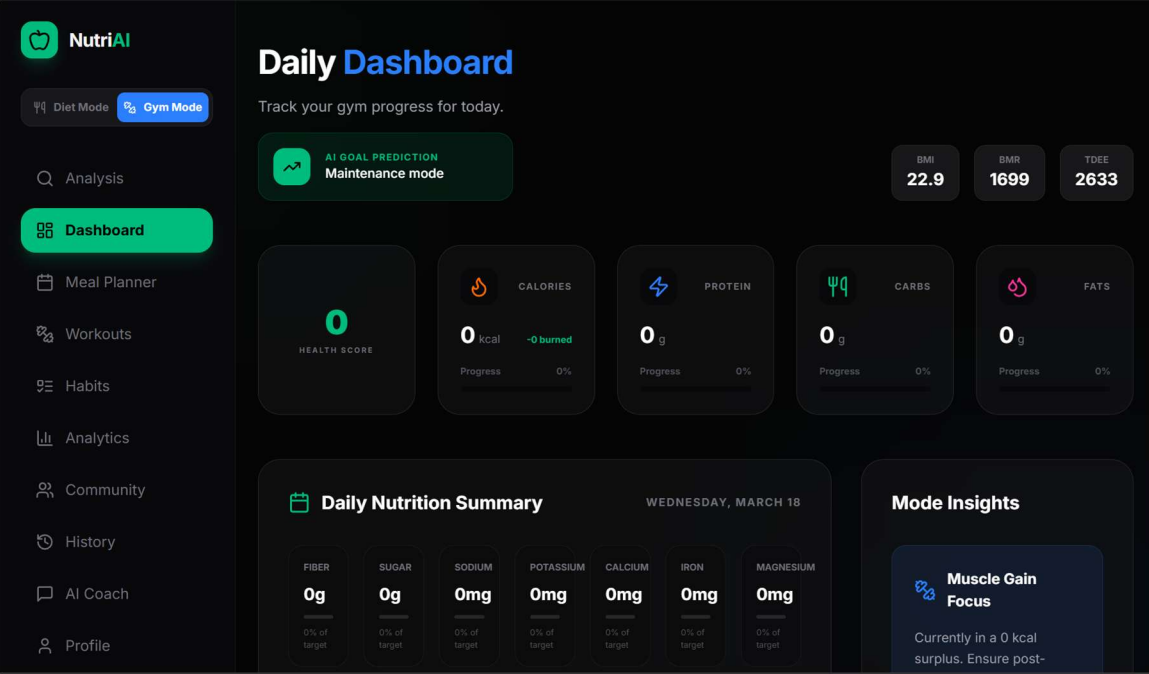
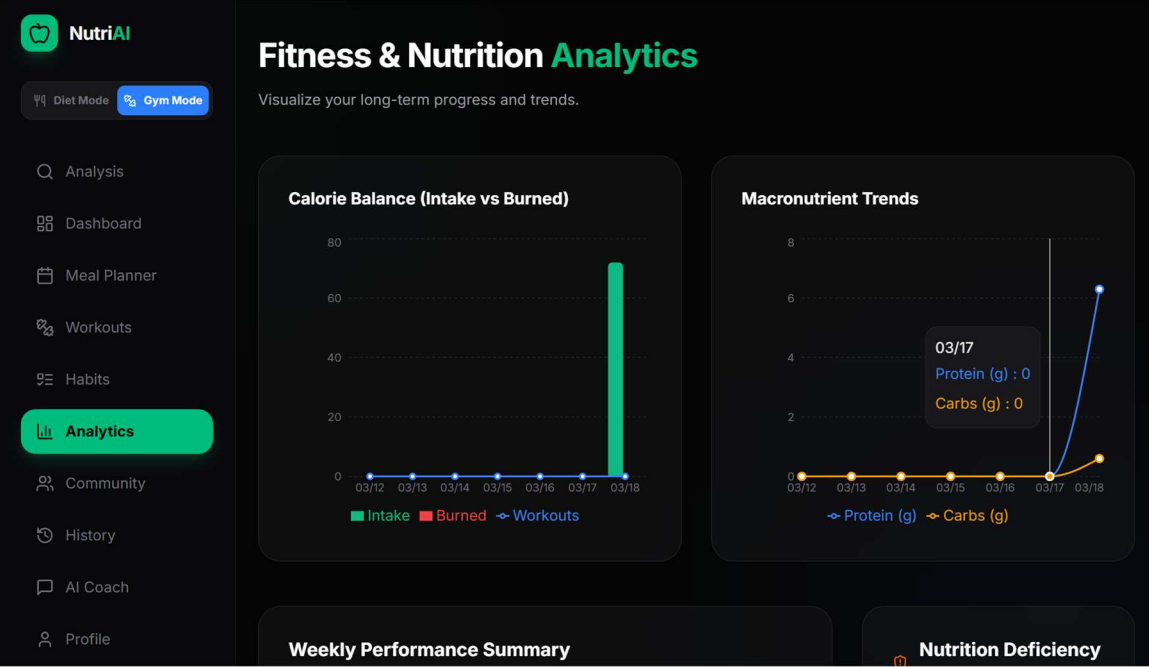
Nutrition AI Analyzer

AI-Powered Food Intelligence System using Google AI Studio

Overview

The Nutrition AI Analyzer is an intelligent system designed to analyze food items and provide detailed nutritional insights, health recommendations, and personalized dietary feedback using advanced Generative AI.





Problem Statement

Users struggle to understand nutritional composition, make informed dietary decisions, and access personalized health insights instantly. This project solves that using LLMs.

Key Features

- Food Nutrition Analysis
- AI-Powered Health Recommendations
- Real-time Prompt-based Analysis
- Context-Aware Responses
- Explainable AI Output

Tech Stack

AI Model: Google Gemini (Google AI Studio)

Prompt Engineering: Structured prompts

Backend: Node.js / Python

Frontend: React / Web UI

Core AI Implementation

Built using Google AI Studio with Gemini models, prompt tuning, and structured pipelines.

🤖 Core AI Implementation

🔥 Built using **Google AI Studio**

This project is powered by **Google AI Studio**, leveraging:

- **Gemini models** for natural language understanding
- **Prompt tuning techniques** for precise outputs
- **Structured prompt pipelines** for consistency

🛠️ Prompt Engineering Strategy

Example Prompt Structure:

...

Input: "2 eggs, 1 slice bread, 1 banana"

Step 1: Identify each food item

Step 2: Retrieve nutritional values

Step 3: Calculate total calories and macros

Step 4: Analyze health impact

Step 5: Suggest improvements

Output: Structured nutritional report + recommendations

...

📊 Sample Output

...

Total Calories: 350 kcal

Protein: 18g | Carbs: 40g | Fats: 12g

Analysis:

- Good protein intake
- Moderate carbs
- Slightly high fat

Recommendation:

- Add fiber (vegetables)
- Reduce processed fats

🛠️ How to Run

```
git clone https://github.com/your-username/nutrition-ai-analyzer
```

```
npm install
```

```
npm start
```

🌍 Future Enhancements

- Image-based food recognition
- User-specific diet personalization
- Integration with health APIs
- Dashboard visualization

🏁 Conclusion

This project demonstrates strong capabilities in:

- AI system design

- Prompt engineering
- Real-world problem solving